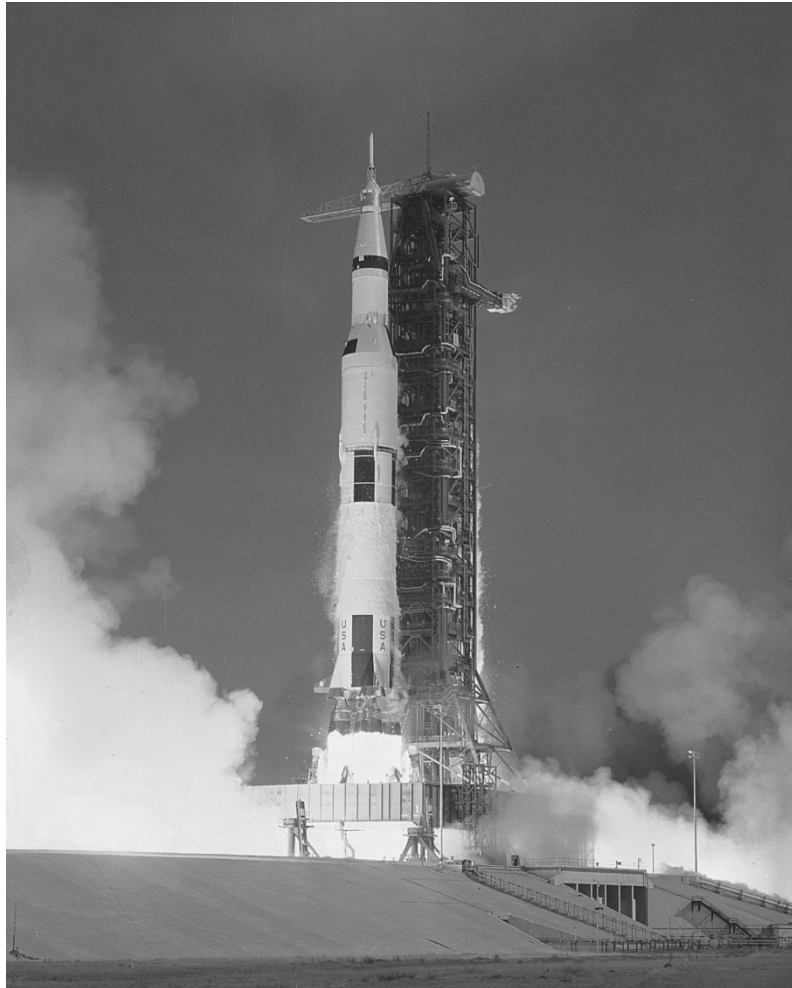


Chapter 3 Integral Relations for a Control Volume



Learning Objectives

- Recognize basic laws of fluid mechanics.
- Understand the Reynolds transport theorem and its role in deriving the basic control volume relations of fluid mechanics.
- Apply conservation of mass relation for fluid flows.
- Calculate velocity, pressure, and force for fluid flows using the linear momentum relation.
- Understand Bernoulli equation and its limitation in calculating momentum and energy.
- Apply the energy conservation equation to a fluid system including pump and turbine.

-
- **Fundamental Physics Laws**
 - **Control Volume, Volume Flow Rate, and Mass Flow Rate**
 - **Reynolds Transport Theorem**
 - **Conservation of Mass**
 - **Conservation of Linear Momentum Equation**
 - **Examples of Force Calculation Using Control Volume**
 - **Bernoulli's Equation**
 - **Conservation of Angular Momentum**
 - **Conservation of Energy Equation**

Basic Physical Laws of Fluid Mechanics

- Three basic approaches to the analysis of fluid mechanics problems:
 1. **Integral, or large-scale analysis in a finite control volume (Chap. 3).**
 2. **Differential, or small-scale analysis in an infinitesimal control volume (Chap. 4).**
 3. **Experimental, or dimensional analysis (Chap. 5).**
- The three approaches are equally important.

Basic Laws of Mechanics for a System

- All the laws of solid mechanics are written for a *system*, defined as an arbitrary quantity of mass of fixed identity.

- **Conservation of mass,**

$$m_{\text{sys}} = \text{const} \quad \text{or} \quad \frac{dm}{dt} = 0$$

- The Newton's second law (**linear momentum** equation),

$$\mathbf{F} = m\mathbf{a} = m \frac{d\mathbf{V}}{dt} = \frac{d(m\mathbf{V})}{dt}$$

- Moment and **angular momentum** relation,

$$\mathbf{M} = \frac{d\mathbf{H}}{dt} \quad \text{where the angular momentum } \mathbf{H} = \sum (\mathbf{r} \times \mathbf{V}) \delta m$$

- **Conservation of energy** (1st law of thermodynamics)

$$\delta Q - \delta W = dE \quad \text{or} \quad \dot{Q} - \dot{W} = \frac{dE}{dt}$$

Basic Laws of Mechanics for a System

- The second law of thermal dynamics,

$$dS \geq \frac{\delta Q}{T} \quad \left(\begin{array}{l} \text{The total entropy of an isolated system can never decrease} \\ \text{over time} \end{array} \right)$$

- State relations: **state relation** (or equation of state) expresses one thermodynamic property as a function of two other independent properties $p = p(\rho, T)$.

$$\text{Ideal gas law: } p = \rho PT$$

Ideal gas is assumed to

- Have negligible volume
 - Have no intermolecular forces
 - Undergo perfectly elastic collision
- Energy-temperature relations with assumption of a calorically perfect gas

$$u = c_v T \quad \text{and} \quad h = c_p T$$

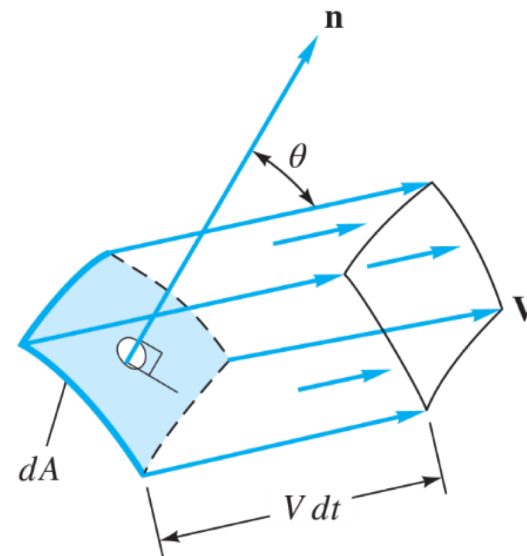
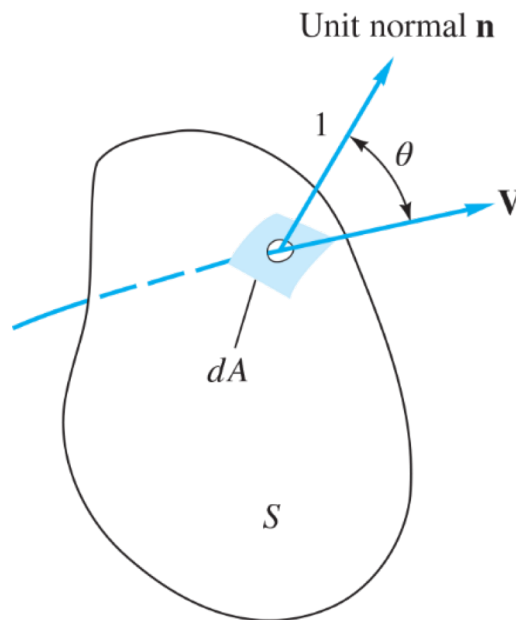
$$\gamma = \frac{c_p}{c_v}$$

Volume and Mass Rate of Flow

- Volume rate of flow (volume flow rate) is defined as the amount of fluid swept through dA in time dt (m^3/s),

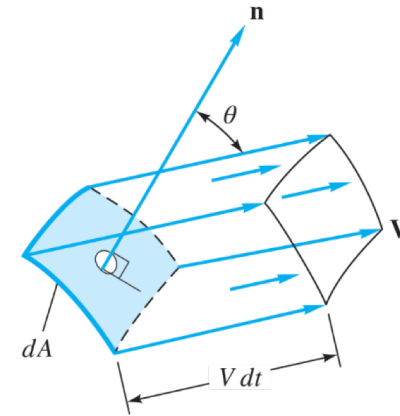
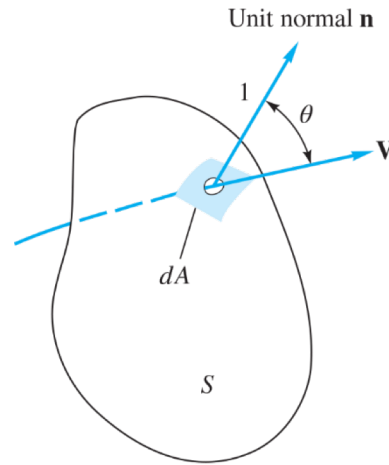
$$Q = \int (\mathbf{V} \cdot \mathbf{n}) dA = \int V_n dA$$

where $\mathbf{n}$ is unit vector normal to dA , positive when it is outward normal unit vector.



Volume and Mass Rate of Flow

$$Q = \int (\mathbf{V} \cdot \mathbf{n}) dA = \int V_n dA$$



- Volume flow can be multiplied by density to obtain mass rate of flow,

$$\dot{m} = \int \rho (\mathbf{V} \cdot \mathbf{n}) dA = \int \rho V_n dA$$

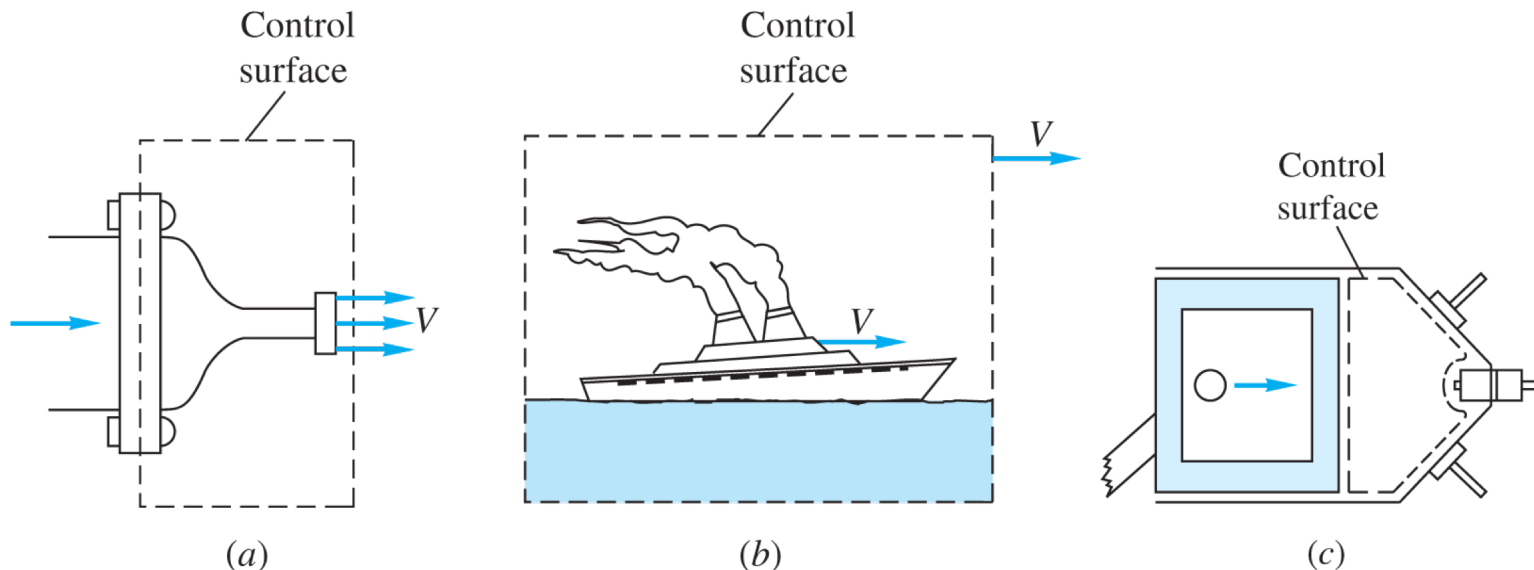
- If density and velocity are constant over the surface S ,

$$\dot{m} = \rho Q = \rho A V$$

- Units for Q are m^3/s , and for $\dot{m}$ are kg/s .

The Reynolds Transport Theorem

- *Reynolds transport theorem* that mathematically connects the **Lagrangian perspective** (tracking a specific, moving *system*) and the **Eulerian perspective** (observing a fixed, stationary region in a *control volume*).
- The desired conversion formula differs slightly according to whether the control volume is fixed, moving, or deformable.



-
- System at time $t + dt$
- System at time t
- Fixed control volume CV
- dA
- θ
- $\mathbf{V}_{out}$
- $\mathbf{n}$, Unit outward normal to dA
- $\mathbf{V}_{in}$
- θ
- $\mathbf{n}$, Unit outward normal to dA
- Arbitrary fixed control surface CS
- $$d\mathcal{V}_{in} = V_{in} dA_{in} \cos \theta_{in} dt = -\mathbf{V} \cdot \mathbf{n} dA dt$$
- $$d\mathcal{V}_{out} = V_{out} dA_{out} \cos \theta_{out} dt = \mathbf{V} \cdot \mathbf{n} dA dt$$

Fixed, Moving and Deformable Control Volume

- Let B be any property of the fluid (mass, momentum, energy, etc.), and β is the amount of B per unit mass.

$$B_{CV} = \int_{CV} \beta dm = \int_{CV} \beta \rho dV \quad \text{and} \quad \beta = \frac{dB}{dm}$$

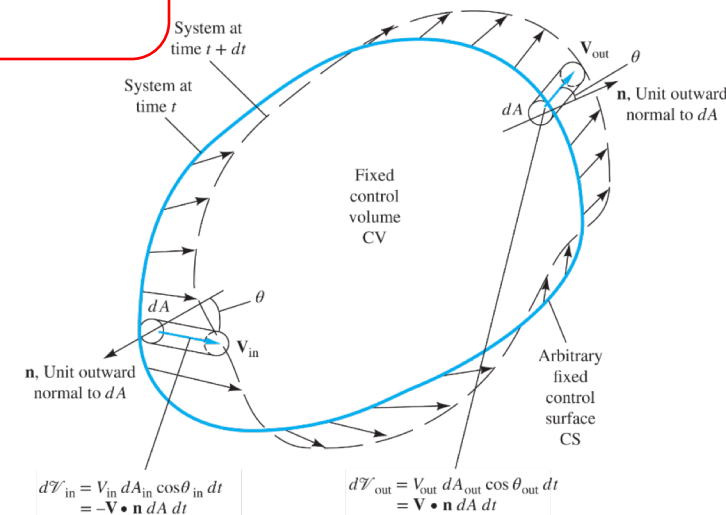
- The compact form of the Reynolds transport theorem is

$$\frac{dB_{sys}}{dt} = \frac{d}{dt} \left(\int_{CV} \beta \rho dV \right) + \int_{CS} \beta \rho (\mathbf{V} \cdot \mathbf{n}) dA$$

- The flow terms of β from the CV are

$$\int_{CS} \beta \rho V \cos \theta dA_{out} - \int_{CS} \beta \rho V \cos \theta dA_{in}$$

Outflow of β from the CV
Inflow of β into the CV



Fixed, Moving and Deformable Control Volume

- Reynolds transport theorem for an arbitrary fixed control volume

$$\frac{dB_{\text{sys}}}{dt} = \frac{d}{dt} \left(\int_{\text{CV}} \beta \rho dV \right) + \int_{\text{CS}} \beta \rho V \cos \theta dA_{\text{out}} - \int_{\text{CS}} \beta \rho V \cos \theta dA_{\text{in}}$$

Outflow Inflow

- Moving or deformable CV at $\mathbf{V}_s$,

$$\frac{dB_{\text{sys}}}{dt} = \frac{d}{dt} \left(\int_{\text{CV}} \beta \rho dV \right) + \int_{\text{CS}} \beta \rho (\mathbf{V}_r \cdot \mathbf{n}) dA$$

where relative velocity $\mathbf{V}_r$ is $\mathbf{V}_r = \mathbf{V} - \mathbf{V}_s$

For a fixed control volume, the volume elements do not vary, then

$$\frac{d}{dt} \left(\int_{\text{CV}} \beta \rho dV \right) = \int_{\text{CV}} \frac{\partial}{\partial t} (\beta \rho) dV$$

Conservation of Mass

- **Conservation of mass:**

For the Reynolds transport theorem, $B = m, \beta = 1$, for a fixed CV

$$\frac{dB_{\text{sys}}}{dt} = \frac{d}{dt} \left(\int_{\text{CV}} \beta \rho dV \right) + \int_{\text{CS}} \beta \rho (\mathbf{V} \cdot \mathbf{n}) dA$$

$$\frac{dB_{\text{sys}}}{dt} = 0 = \frac{d}{dt} \left(\int_{\text{CV}} \rho dV \right) + \int_{\text{CS}} \rho (\mathbf{V} \cdot \mathbf{n}) dA$$

$$\int_{\text{CV}} \frac{\partial \rho}{\partial t} dV + \int_{\text{CS}} \rho (\mathbf{V} \cdot \mathbf{n}) dA = 0$$

Conservation of Mass

- One-dimensional inlets and outlets

$$\int_{\text{CV}} \frac{\partial \rho}{\partial t} dV + \sum_i (\rho_i A_i V_i)_{\text{out}} - \sum_i (\rho_i A_i V_i)_{\text{in}} = 0$$

- One-dimensional and Steady flow, $\partial \rho / \partial t = 0$

$$\sum_i (\rho_i A_i V_i)_{\text{in}} = \sum_i (\rho_i A_i V_i)_{\text{out}}$$

Inflow Outflow

- One-dimensional, steady with one inlet and one outlet,

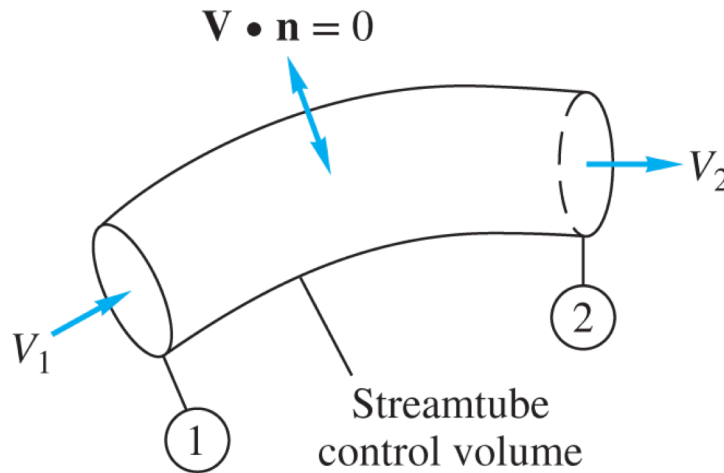
$$\rho_1 A_1 V_1 = \rho_2 A_2 V_2 \quad \text{or} \quad \dot{m}_1 = \dot{m}_2$$

- One-dimensional, steady, Incompressible with one inlet/outlet $\rho = \text{const}$,

$$A_1 V_1 = A_2 V_2 \quad \text{or} \quad Q_1 = Q_2$$

Conservation of Mass

Example: From the figure find exit velocity V_2 for steady flow through a stream tube, A_1 , A_2 , and V_1 are given.



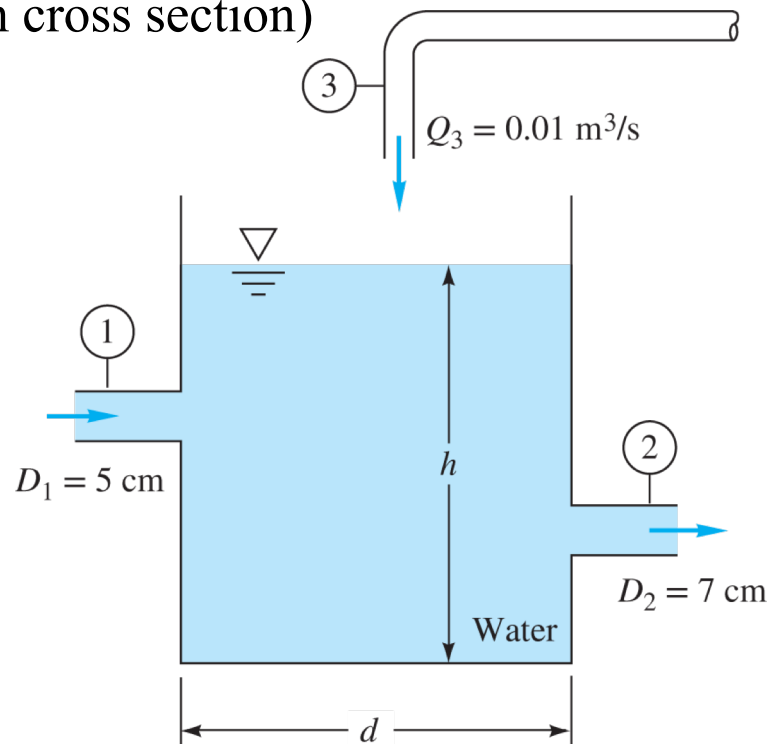
Solution: From conservation of mass,

$$A_1 V_1 = A_2 V_2 \quad \Rightarrow \quad V_2 = \frac{A_1}{A_2} V_1$$

Conservation of Mass

Example: The open tank contains water at 20°C and is being filled through sections 1 and 3. (1) Derive an expression for water level change dh/dt in terms of Q_1, Q_2, Q_3 , and d ; (2) for constant water level h , determine V_2 if $V_1 = 3$ m/s and $Q_3 = 0.01$ m³/s.

(Assume the tank is circular in cross section)



Conservation of Mass

Solution:

Select a CV enclosing water in the tank

Applying the mass conservation eq.

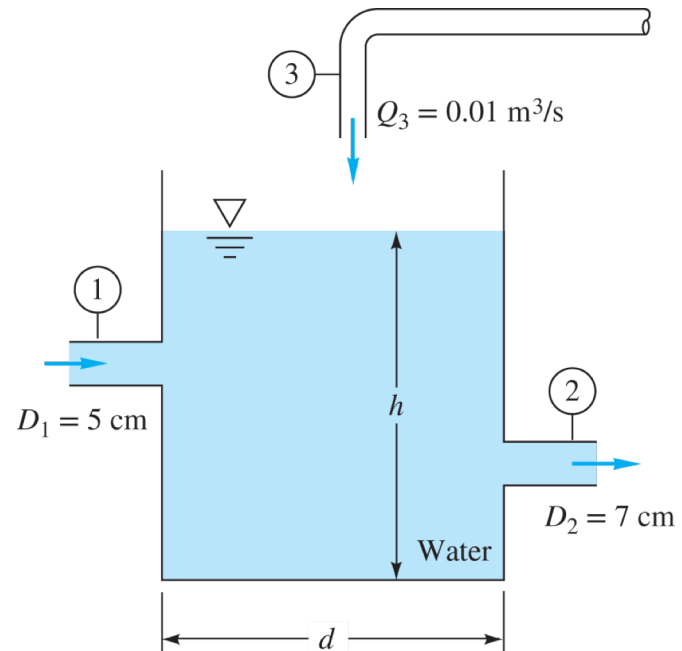
$$\frac{d}{dt} \left(\int_{CV} \rho dV \right) + \sum_i (\rho_i A_i V_i)_{out} - \sum_i (\rho_i A_i V_i)_{in} = 0$$

$$= \rho \frac{\pi d^2}{4} \frac{dh}{dt} + \rho (Q_2 - Q_1 - Q_3) = 0$$

$$\Rightarrow \frac{dh}{dt} = \frac{(Q_2 - Q_1 - Q_3)}{\pi d^2 / 4}$$

If h is constant, then, $Q_1 + Q_3 = Q_2 = \frac{\pi d^2}{4} V_2$

$$\frac{\pi (0.05)^2}{4} (3) + 0.01 = \frac{\pi (0.07)^2}{4} V_2 \Rightarrow V_2 = 4.13 \text{ m/s}$$



-
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The Linear Momentum Equation

- The linear momentum: $\mathbf{B} = m\mathbf{V}$, $\beta = d\mathbf{B}/dm = \mathbf{V}$, fixed CV

$$\left(\frac{d\mathbf{B}_{\text{sys}}}{dt} = \frac{d}{dt} \left(\int_{\text{CV}} \beta \rho dV \right) + \int_{\text{CS}} \beta \rho (\mathbf{V} \cdot \mathbf{n}) dA \right)$$

$$\frac{d(m\mathbf{V})_{\text{sys}}}{dt} = \sum \mathbf{F} = \frac{d}{dt} \left(\int_{\text{CV}} \mathbf{V} \rho dV \right) + \int_{\text{CS}} \mathbf{V} \rho (\mathbf{V} \cdot \mathbf{n}) dA$$

$$\text{or } \sum \mathbf{F} = \frac{d}{dt} \left(\int_{\text{CV}} \mathbf{V} \rho dV \right) + \int_{\text{CS}} \mathbf{V} \rho (\mathbf{V} \cdot \mathbf{n}) dA$$

- Force term $\sum \mathbf{F}$ should include all the surface forces (pressure, shear stress), volumetric body forces (gravity, buoyancy), and external forces.

The Linear Momentum Equation

- 1-D Momentum flux at inlets and outlets

$$\sum \mathbf{F} = \frac{d}{dt} \left(\int_{CV} \mathbf{V} \rho dV \right) + \sum_i (\dot{m}_i \mathbf{V}_i)_{\text{out}} - \sum_i (\dot{m}_i \mathbf{V}_i)_{\text{in}}$$

Momentum flow

- Steady and nondeforming flow,

$$\sum \mathbf{F} = \sum_i (\dot{m}_i \mathbf{V}_i)_{\text{out}} - \sum_i (\dot{m}_i \mathbf{V}_i)_{\text{in}}$$

- One inlet/outlet (1-inlet, 1-outlet, $\dot{m}_1 = \dot{m}_2 = \dot{m}$)

$$\sum \mathbf{F} = \dot{m} (\mathbf{V}_2 - \mathbf{V}_1)$$

Pressure Condition at a Jet Exit

- External pressure force on a closed control surface (CS) is normal to the surface and inward, for the two cases shown

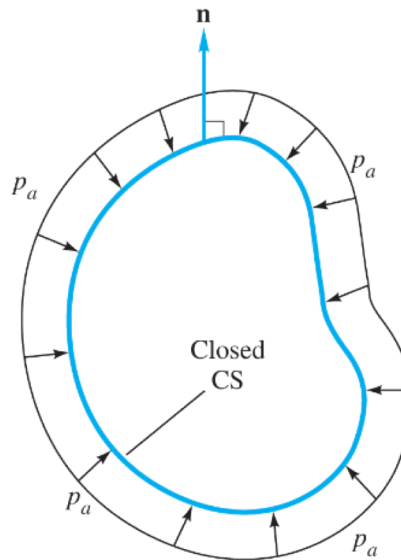
(a) uniform pressure

$$\mathbf{F} = -p_a \int \mathbf{n} dA \equiv 0$$

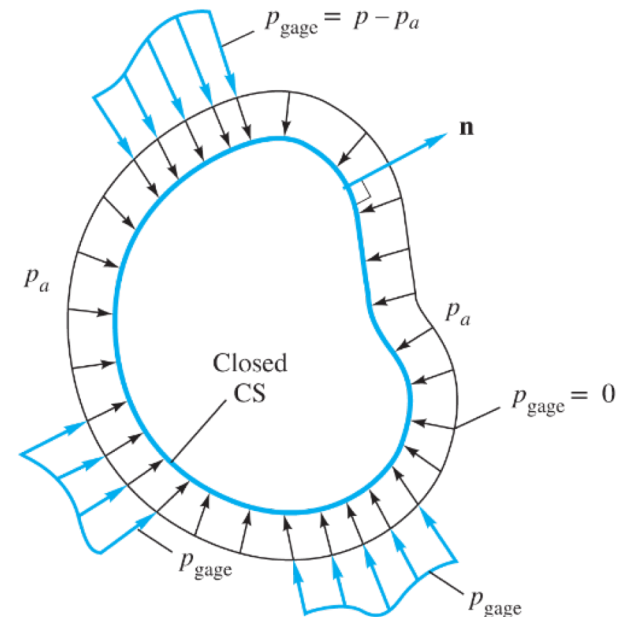
(b) nonuniform pressure

$$\mathbf{F} = \int (p - p_a)(-\mathbf{n}) dA \equiv 0$$

➔
$$\mathbf{F} = \int p_{gage}(-\mathbf{n}) dA \equiv 0$$



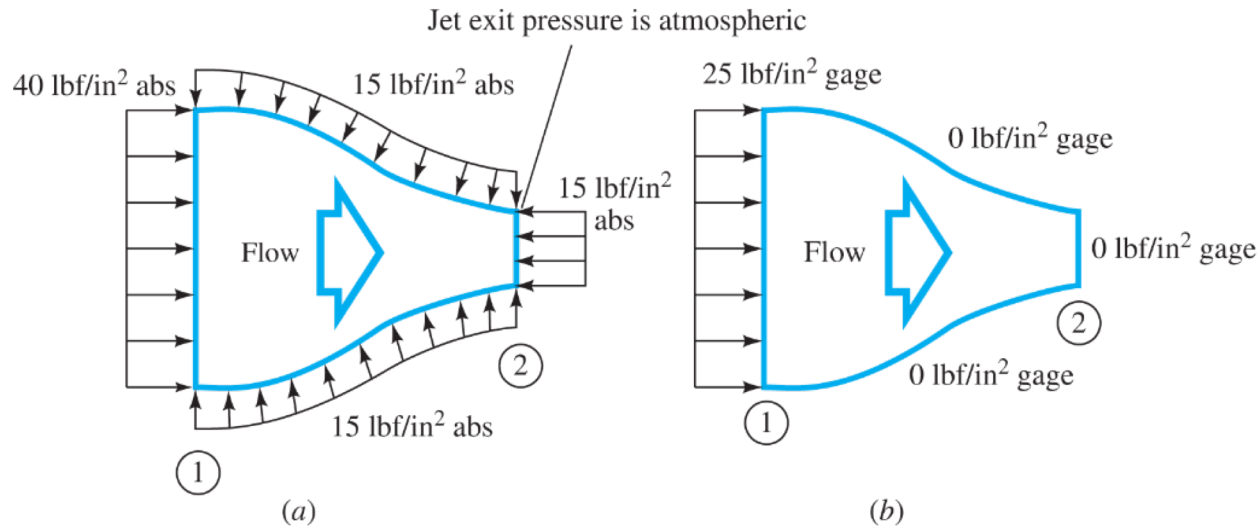
(a)



(b)

Pressure Condition at a Jet Exit

Example: For a nozzle section shown, find the net pressure if $D_1 = 3$ in.



Solution: Select the outside of the nozzle plus the cut sections (1) and (2) as CV,

$$\mathbf{F} = p_{\text{gage},1} (-\mathbf{n}) A_1$$

$$\mathbf{F} = (40 - 15) \frac{\text{lbf}}{\text{in}^2} \mathbf{i} \frac{\pi}{4} (3 \text{ in})^2$$

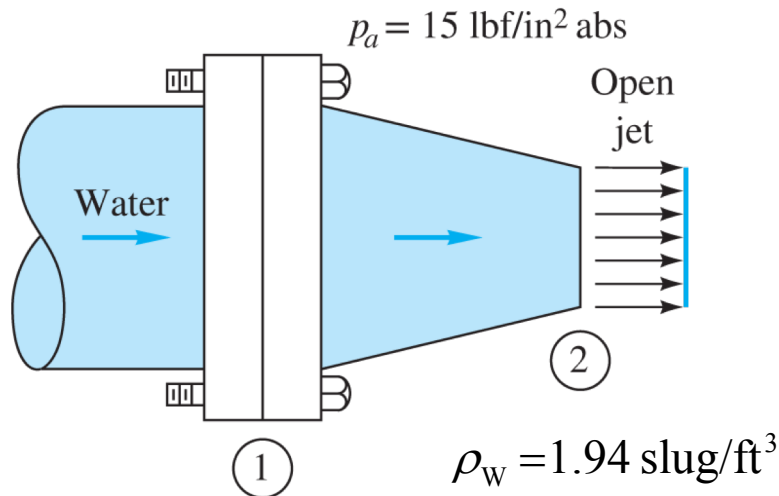
$$\rightarrow \mathbf{F} = 177 \mathbf{i} \text{ lbf}$$

Anchoring Force to Hold an Elbow or Vane Stationary

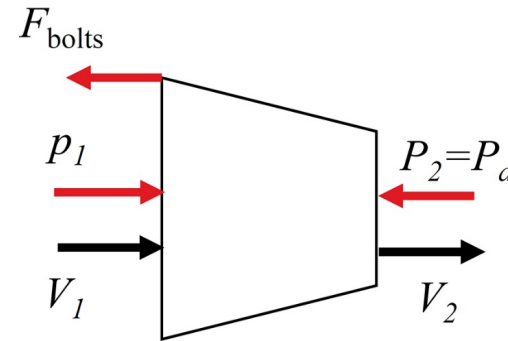
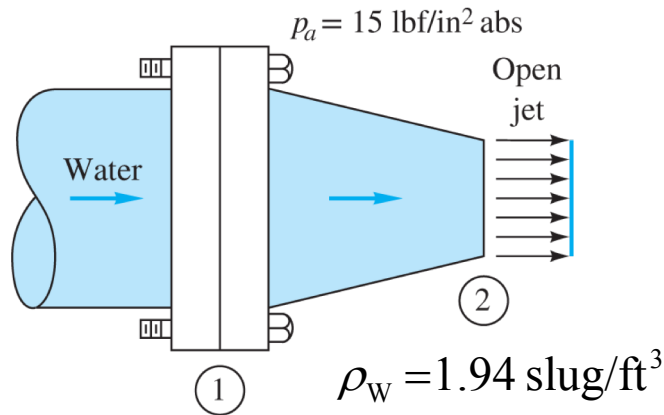
- When a liquid flow passes through an elbow that changes the direction of flow, or water jet deflected by a vane into different direction, it is crucial to determine the anchoring forces required to hold the elbow and vane stationary.

Example: The horizontal nozzle has $D_1 = 12$ in and $D_2 = 6$ in, with inlet pressure $p_1 = 38$ lbf/in² absolute and $V_2 = 56$ ft/s.

For water at 20°C, determine the horizontal force provided by the flange bolts.



Anchoring Force to Hold an Elbow or Vane Stationary



Solution: Select the outside of the nozzle plus the cut sections (1) and (2) as CV. Applying the mass balance eq. $A_1 V_1 = A_2 V_2$

$$V_1 \frac{\pi (12)^2}{4} = (56) \frac{\pi (6)^2}{4} \quad \Rightarrow \quad V_1 = 14 \text{ ft/s}$$

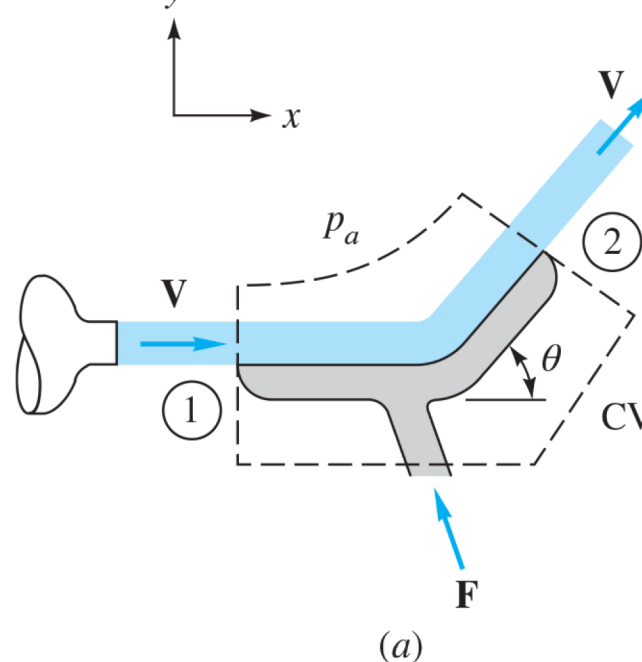
Applying the linear momentum eq., $\sum \mathbf{F} = \dot{m}(\mathbf{V}_2 - \mathbf{V}_1)$

$$F_x = -F_{\text{bolts}} + (23) \frac{\pi (12)^2}{4} = (1.94) \frac{\pi}{4} \left(\frac{12}{12} \right)^2 (14)(56 - 14) \quad \Rightarrow \quad F_{\text{bolts}} = 1700 \text{ lbf}$$

Anchoring Force to Hold an Elbow or Vane Stationary

Example: A fixed vane turns a water jet of area A through an angle θ without changing its velocity magnitude V . The flow is steady, pressure is p_a everywhere. (a) Find the components F_x and F_y of the applied vane force. (b) Express the force F and the angle ϕ between F and the horizontal in term of θ .

(Also, ignore the weight of fluid and the vane weight within the CV.)



Anchoring Force to Hold an Elbow or Vane Stationary

Solution:

(a) **Select a CV cuts through the inlet and exit of the jet through the vane support** --- vane friction is self-canceling.

By the mass conservation,

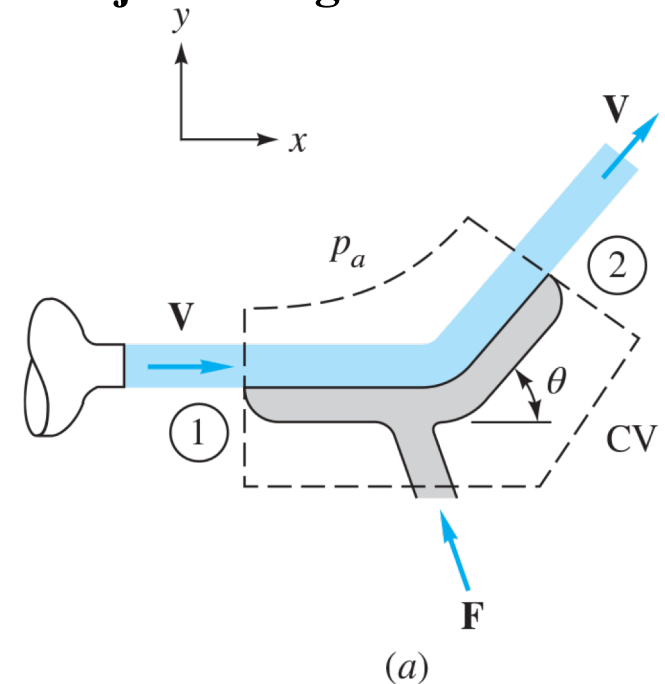
$$\dot{m}_1 = \dot{m}_2 = \dot{m}$$

Applying the momentum eq. and neglecting weight of fluid and vane

$$\mathbf{F} = \dot{m}_2 \mathbf{V}_2 - \dot{m}_1 \mathbf{V}_1 = \dot{m} (\mathbf{V}_2 - \mathbf{V}_1)$$

Components of the force in x and y direction are:

$$F_x = \dot{m}V (\cos \theta - 1) \quad F_y = \dot{m}V \sin \theta \quad \text{where } \dot{m}V = \rho AV^2$$



Anchoring Force to Hold an Elbow or Vane Stationary

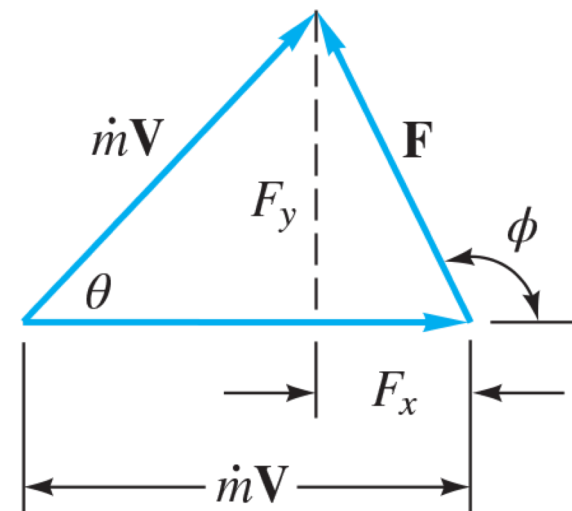
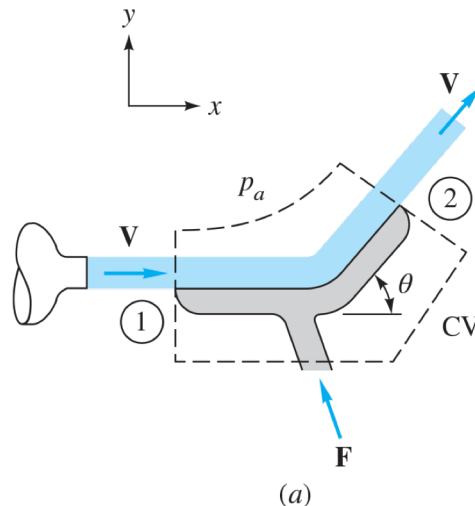
Solution (cont.):

(b) The resultant force magnitude is

$$F = \sqrt{F_x^2 + F_y^2} = \dot{m}V \sqrt{\sin^2 \theta + (\cos \theta - 1)^2} = \dot{m}V \sqrt{2(1 - \cos \theta)} = 2\dot{m}V \sin \frac{\theta}{2}$$

From the geometry,

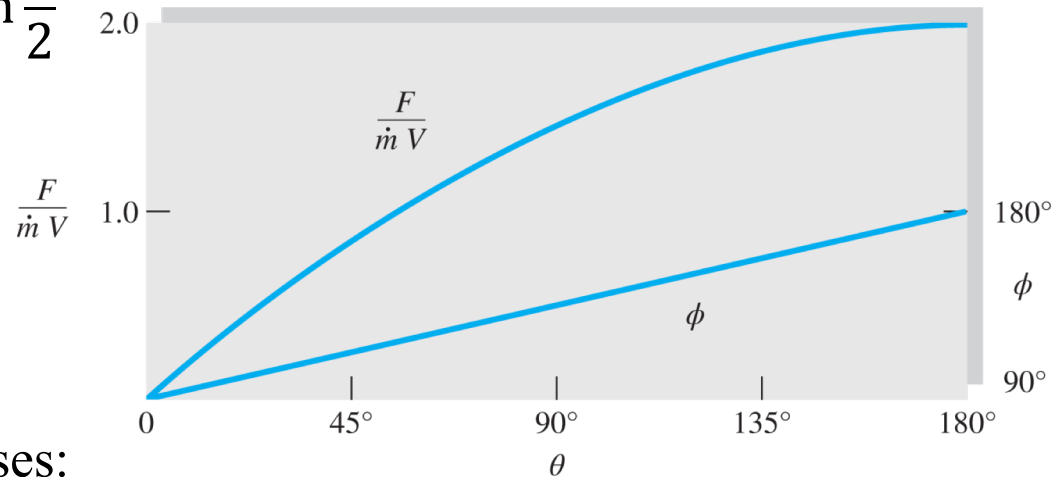
$$\phi = 180^\circ - \tan^{-1} \frac{F_y}{F_x} = 90^\circ + \frac{\theta}{2}$$



Anchoring Force to Hold an Elbow or Vane Stationary

- Relation between F and θ

$$\frac{F}{\dot{m}V} = 2 \sin \frac{\theta}{2}$$



Two special cases:

- At $\theta = 180^\circ$, maximum force occurs.
The jet is turned around and thrown back in the opposite direction with its momentum reversed.
- At very small θ , we have $F = \dot{m}V\theta$, and $\phi = 90^\circ$
The force is linearly proportional to the turning angle, and acts normal to the jet (lift force).

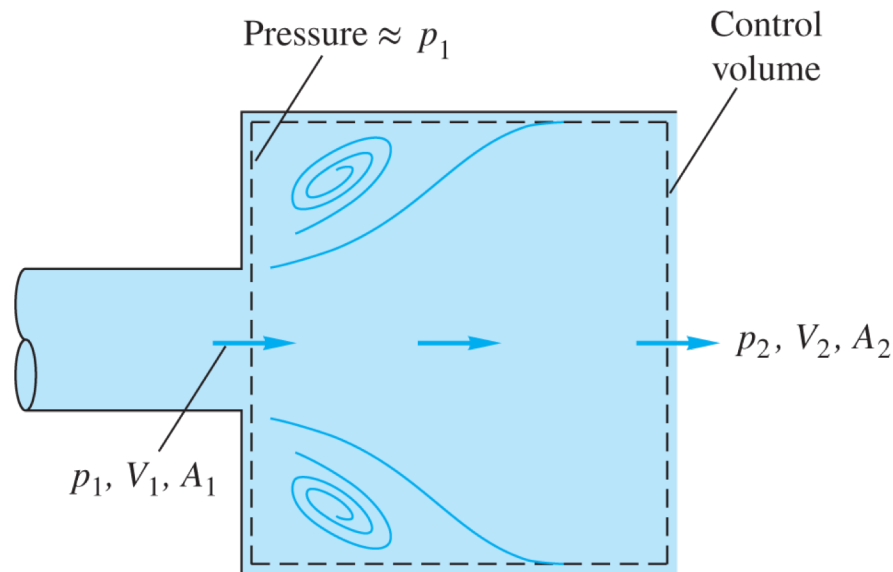
Friction Force Exerted by Solid Wall on Fluid Flow

- Many engineering applications need to find the friction force exerted by pipe wall on the fluid flow in a pipe or determine the drag force induced by a flat plate on the flow over it.

Example: A pipe flow suddenly expands from A_1 to A_2 .

Assume the flow is steady and incompressible, and $p_w \approx p_1$ on the corner annular ring as shown.

p_1, V_1, A_1, p_2, A_2 , and ρ are given, find V_2 and the wall friction.



Friction Force Exerted by Solid Wall on Fluid Flow

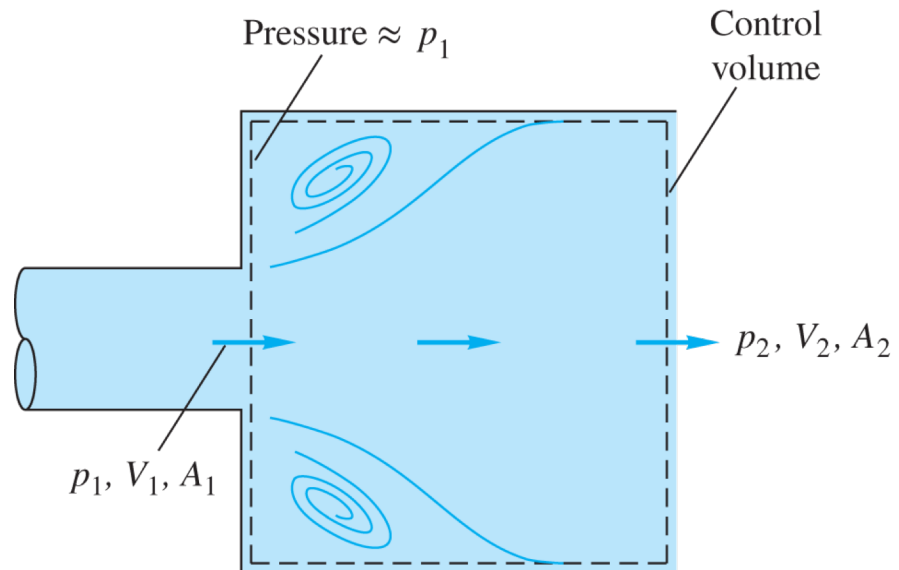
Solution:

Select the inside of the expansion plus the cut sections (1) and (2) as CV

From the mass conservation,

$$A_1 V_1 = A_2 V_2$$

$$V_2 = \frac{A_1}{A_2} V_1$$



Friction Force Exerted by Solid Wall on Fluid Flow

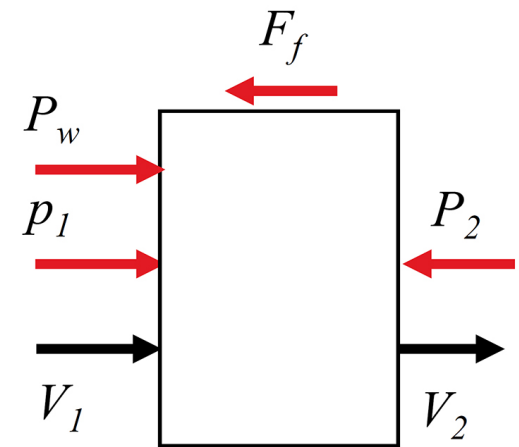
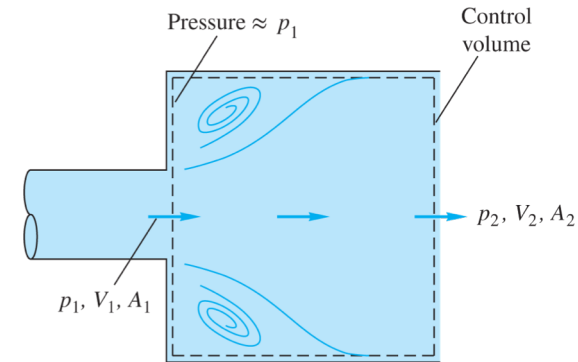
Solution (Con'd):

Applying the linear momentum eq.

$$\sum \mathbf{F} = \dot{m}(\mathbf{V}_2 - \mathbf{V}_1)$$

$$F_x = p_1 A_1 + p_w (A_2 - A_1) - p_2 A_2 - F_f = \rho A_1 V_1 (V_2 - V_1)$$

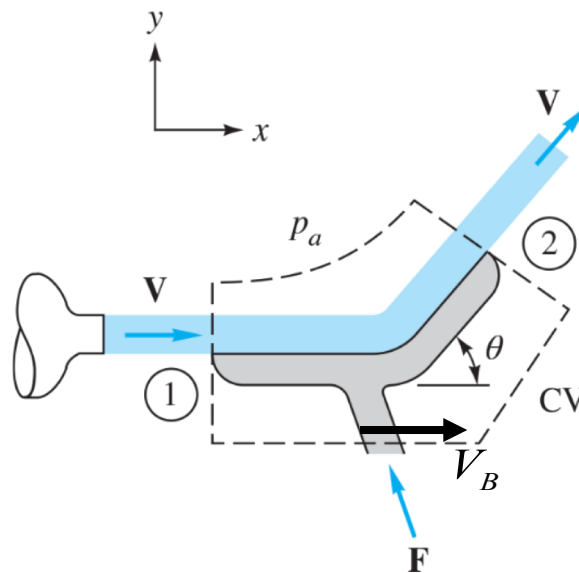
$$\Rightarrow F_f = A_2 (p_1 - p_2) + \rho A_1 V_1^2 \left(1 - \frac{A_1}{A_2} \right)$$



Moving Control Volume

- A non-deforming control volume that translates at constant speed is inertial because there is no acceleration. In this case, the linear momentum equation is still applicable, except that the velocity V should be treated as the relative velocity.

Example 1: Re-consider the previous example assuming the vane is moving at a constant velocity V_B , find the components F_x and F_y of the applied vane force.



Moving Control Volume

Solution:

Select a CV cuts through the inlet and exit of the jet through the vane support

Relative velocity relation, $\mathbf{V}_A = \mathbf{V}_B + \mathbf{V}_{A/B}$

$$\rightarrow V_r = V_1 - V_B \quad (V_1 \text{ and } V_B \text{ are in the same direction.})$$

By the mass conservation,

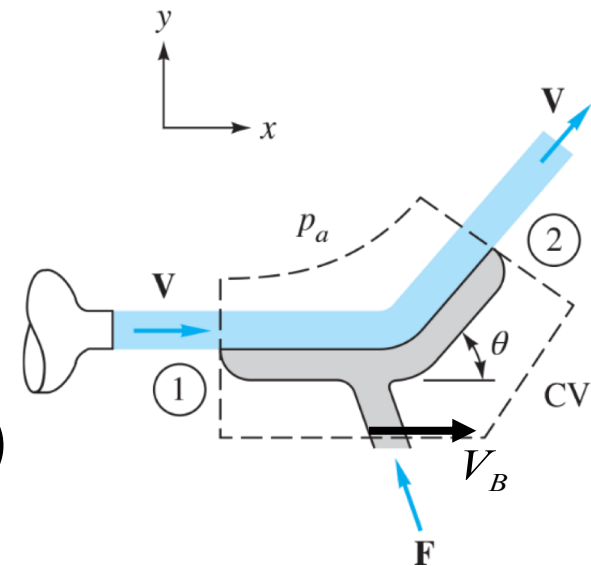
$$\dot{m}_1 = \dot{m}_2 = \dot{m} \quad \text{where} \quad \dot{m} = \rho(V_1 - V_B)A$$

Applying the linear momentum eq.

$$\sum \mathbf{F} = \dot{m}(\mathbf{V}_2 - \mathbf{V}_1)$$

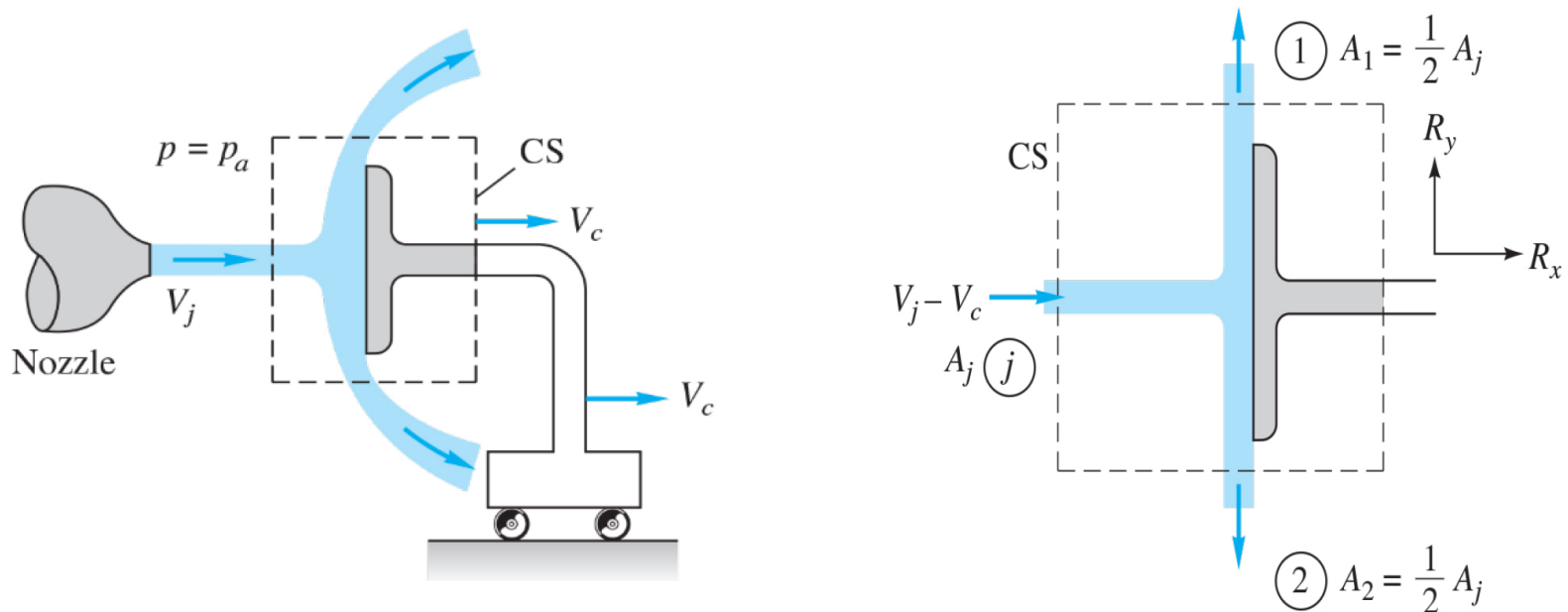
$$\rightarrow F_x = \dot{m}V_r(\cos\theta - 1) = \dot{m}(V_1 - V_B)^2 A(\cos\theta - 1)$$

$$F_y = \dot{m}V_r \sin\theta = \dot{m}(V_1 - V_B)^2 A \sin\theta$$



Moving Control Volume

Example: A water jet of velocity V_j impinges normal to a flat plate that moves at velocity V_c . Find the force required to keep the plate moving at constant velocity if $\rho_j = 1000 \text{ kg/m}^3$, $A_j = 3 \text{ cm}^2$, $V_j = 20 \text{ m/s}$, and $V_c = 15 \text{ m/s}$.



Moving Control Volume

Solution: Select CV cuts through the plate support.

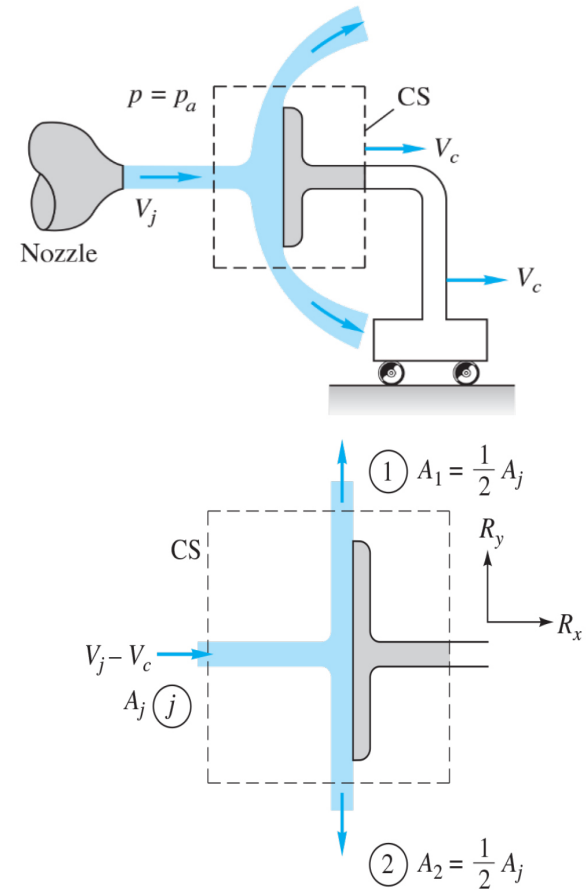
Assumptions:

1. steady flow with respect to the moving plate,
2. negligible weight of jet and plate,
3. symmetric jet,
4. no lateral spreading.

By the mass conservation,

$$\dot{m}_{out} = \dot{m}_{in}$$

$$\rho_1 A_1 V_1 + \rho_2 A_2 V_2 = \rho_j A_j (V_j - V_c)$$



Moving Control Volume

Solution (continued):

Applying the momentum eq. $\mathbf{F} = \sum \dot{m}_{\text{out}} \mathbf{V}_{\text{out}} - \sum \dot{m}_{\text{in}} \mathbf{V}_{\text{in}}$

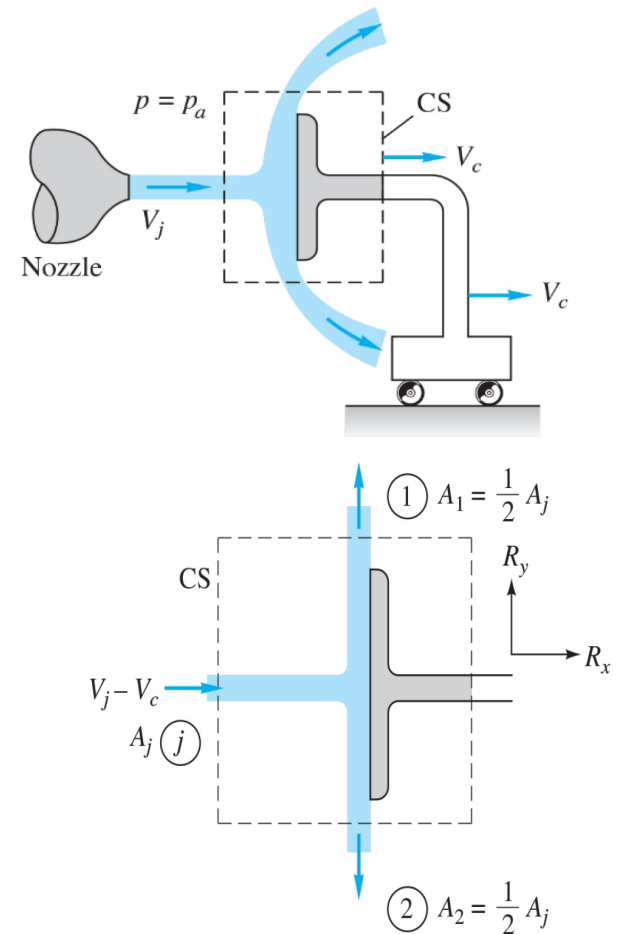
$$V_1 = V_2 = V_j - V_c = 20 - 15 = 5 \text{ m/s}$$

$$F_x = R_x = -\dot{m}_j u_j = -\rho_j A_j (V_j - V_c)^2$$

$$\rightarrow R_x = -(1000)(0.0003)(20 - 15)^2 = -7.5 \text{ N}$$

$$F_y = R_y = \dot{m}_1 v_1 + \dot{m}_2 v_2 = \dot{m}_1 V_1 + \dot{m}_2 (-V_2)$$

$$\rightarrow R_y = 0$$



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- **Fundamental Physics Laws**
 - **Control Volume, Volume Flow Rate, and Mass Flow Rate**
 - **Reynolds Transport Theorem**
 - **Conservation of Mass**
 - **Conservation of Linear Momentum Equation**
 - **Examples of Force Calculation Using Control Volume**
 - **Bernoulli's Equation**
 - **Conservation of Angular Momentum**
 - **Conservation of Energy Equation**

Frictionless Flow: The Bernoulli Equation

- A classic linear momentum analysis between pressure, velocity, and elevation in a **frictionless** flow results in the **Bernoulli equation**.

$$\frac{p_1}{\rho} + \frac{1}{2} V_1^2 + gz_1 = \frac{p_2}{\rho} + \frac{1}{2} V_2^2 + gz_2 = \text{const} \quad (\text{steady form})$$

- The Bernoulli equation can be derived using the following restrictive assumptions:
 1. **Steady flow**: a common assumption for most flows.
 2. **Incompressible flow**: applicable for liquid flow, and gaseous flow with Mach number less than 0.3.
 3. **Frictionless flow**: restrictive---solid walls and mixing introduce friction.
 4. **Flow along a single streamline**: different streamlines may have different Bernoulli constants.

Frictionless Flow: The Bernoulli Equation

Conservation of mass equation

$$\frac{d}{dt} \left(\int_{CV} \rho dV \right) + \dot{m}_{out} - \dot{m}_{in} = 0 \approx \frac{\partial \rho}{\partial t} + d\dot{m}$$

$$d\dot{m} = d(\rho AV) = -\frac{\partial \rho}{\partial t} A ds$$

Linear momentum equation in the streamwise direction

$$\sum dF_s = \frac{d}{dt} \left(\int_{CV} \rho dV \right) + (\dot{m}V)_{out} - (\dot{m}V)_{in} \approx \frac{\partial}{\partial t} (\rho V) A ds + d(\dot{m}V)$$

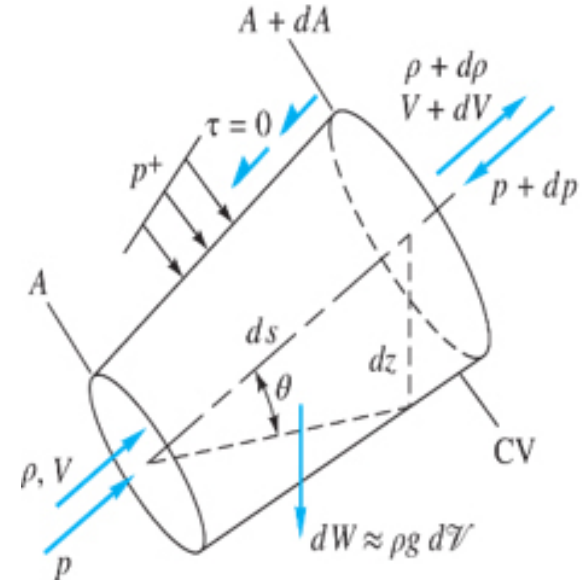
$V_s = V$ because s is the streamline direction

Neglecting shear force on the wall, only pressure and gravity forces remain.

$$dF_{s,grav} = -dW \sin \theta = -\gamma A ds \sin \theta = -\gamma A dz$$

$$dF_{s,press} = \frac{1}{2} dp dA - dp(A + dA) \approx -A dp$$

Elemental fixed streamtube CV



Frictionless Flow: The Bernoulli Equation

Linear momentum equation is rearranged as

$$\begin{aligned}\sum dF_s &= -\gamma Adz - Adp = \frac{\partial}{\partial t}(\rho V)Ads + d(\dot{m}V) \\ &= \cancel{\frac{\partial \rho}{\partial t} V} Ads + \frac{\partial V}{\partial t} \rho Ads + \dot{m} dV + \cancel{V d\dot{m}} \quad (\text{canceled by the continuity relation})\end{aligned}$$

$$\frac{\partial V}{\partial t} ds + \frac{dp}{\rho} + VdV + g dz = 0, \left(\frac{\gamma}{\rho} = g\right)$$

Bernoulli's equation for unsteady frictionless low along a streamline

$$\left| \int_1^2 \frac{\partial V}{\partial t} ds + \int_1^2 \frac{dp}{\rho} + \frac{1}{2} (V_2^2 - V_1^2) + g(z_2 - z_1) = 0 \right|$$

Frictionless Flow: The Bernoulli Equation

For steady incompressible (constant-density) flow,

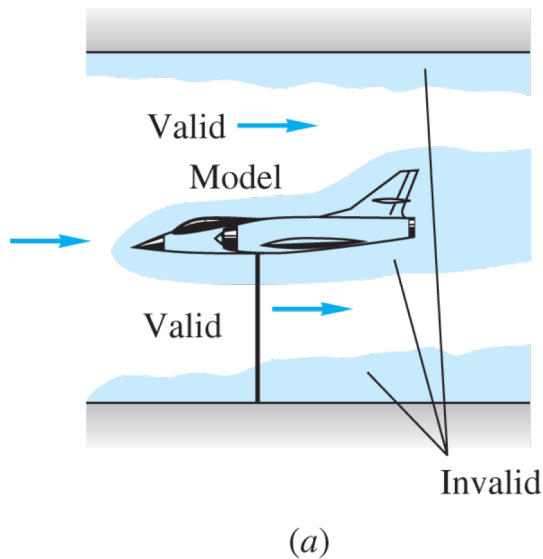
$$\frac{p_2 - p_1}{\rho} + \frac{1}{2}(V_2^2 - V_1^2) + g(z_2 - z_1) = 0$$

$$\text{or} \quad \frac{p_1}{\rho} + \frac{1}{2}V_1^2 + gz_1 = \frac{p_2}{\rho} + \frac{1}{2}V_2^2 + gz_2 = \text{const}$$

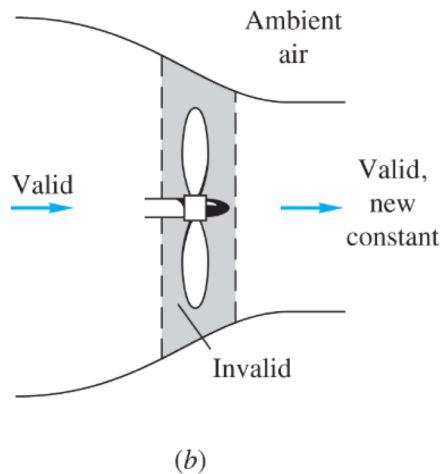
- For frictionless, and incompressible fluid, the Bernoulli equation may also be interpreted as an **idealized energy equation**. The changes from 1 to 2 in above equation represents *reversible* pressure work, kinetic and potential energy changes.

Restrictions on the Bernoulli Equation

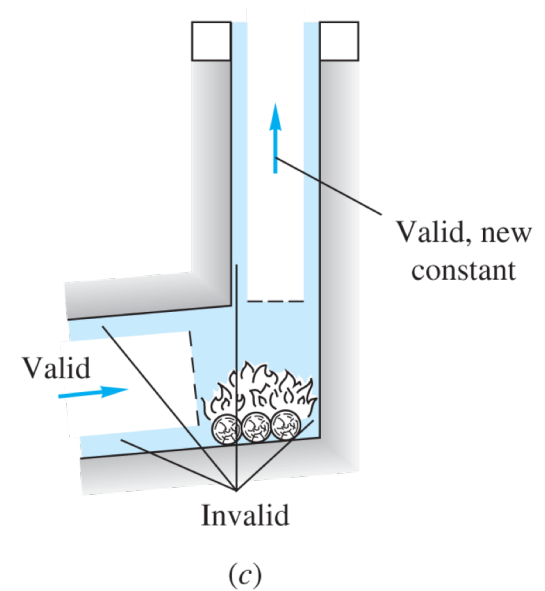
- Regions of validity and invalidity of the Bernoulli equation



Tunnel



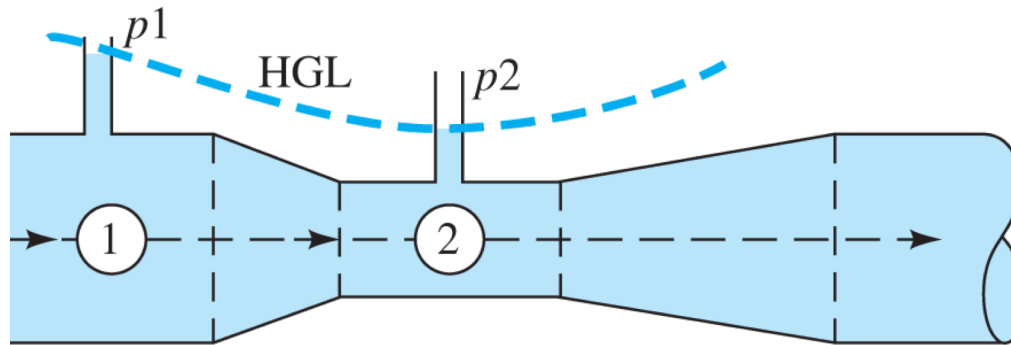
Propeller



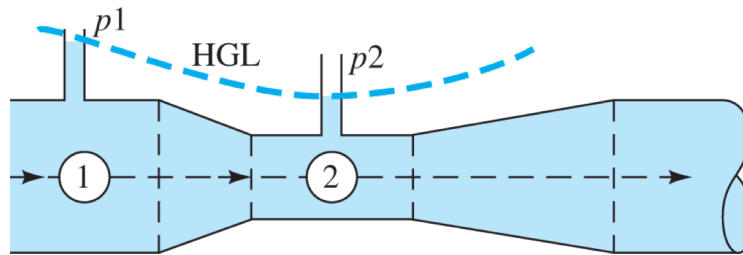
Chimney

Frictionless Flow: The Bernoulli Equation

Example: A constriction in a pipe will cause the velocity to rise and the pressure to fall at section 2 in the throat. The pressure difference is a measure of the flow rate through the pipe. The smoothly necked-down system is called a *venturi tube*. Find an expression for the mass flow in the tube as a function of the pressure change.



Frictionless Flow: The Bernoulli Equation



Solution: Apply Bernoulli eq. along the center streamline from section 1 to 2.

$$\frac{p_1}{\rho} + \frac{1}{2}V_1^2 + gz_1 = \frac{p_2}{\rho} + \frac{1}{2}V_2^2 + gz_2$$

If the tube is horizontal, $z_1 = z_2$ and solve for V_2 ,

$$V_2^2 - V_1^2 = \frac{2\Delta p}{\rho} \quad \Delta p = p_1 - p_2$$

From the mass conservation, $A_1V_1 = A_2V_2$ ➔ $V_1 = \beta^2 V_2$ $\beta = \frac{D_2}{D_1}$

Combining the above equations, $V_2 = \sqrt{\frac{2\Delta p}{\rho(1-\beta^4)}}$

So the mass flow rate is $\dot{m} = \rho A_2 V_2 = \sqrt{\frac{2\rho\Delta p}{(1-\beta^4)}}$

The Angular Momentum Equation

- **The angular momentum:** $\mathbf{B} = \mathbf{H}_0$, the instantaneous angular momentum about O by integration over the mass dm is:

$$\mathbf{H}_0 = \int_{\text{syst}} (\mathbf{r} \times \mathbf{V}) dm \quad \text{so} \quad \beta = \frac{d\mathbf{H}_0}{dm} = \mathbf{r} \times \mathbf{V}$$

- For a nondeformable inertial control volume,

$$\frac{d(\mathbf{H}_0)_{\text{syst}}}{dt} = \sum \mathbf{M}_0 = \frac{d}{dt} \left(\int_{\text{CV}} (\mathbf{r} \times \mathbf{V}) \rho dV \right) + \int_{\text{CS}} (\mathbf{r} \times \mathbf{V}) \rho (\mathbf{V} \cdot \mathbf{n}) dA$$

$$\text{or} \quad \sum \mathbf{M}_0 = \frac{d}{dt} \left(\int_{\text{CV}} (\mathbf{r} \times \mathbf{V}) \rho dV \right) + \int_{\text{CS}} (\mathbf{r} \times \mathbf{V}) \rho (\mathbf{V} \cdot \mathbf{n}) dA$$

- Steady flow with 1-D inlets and outlets

$$\sum \mathbf{M}_0 = \sum_i (\mathbf{r}_i \times \mathbf{V}_i)_{\text{out}} \dot{m}_{i,\text{out}} - \sum_i (\mathbf{r}_i \times \mathbf{V}_i)_{\text{in}} \dot{m}_{i,\text{in}}$$

Angular momentum flow

The Energy Equation

- **Conservation of energy** is based on the 1st law of thermodynamics
- Conservation of energy: $B = E$, $\beta = dE/dm = e$, fixed CV

$$\left(\frac{dB_{\text{sys}}}{dt} = \frac{d}{dt} \left(\int_{\text{CV}} \beta \rho dV \right) + \int_{\text{CS}} \beta \rho (\mathbf{V} \cdot \mathbf{n}) dA \right)$$

(Net flux of heat into the CV) - (Rate of work done by the CV)
= (Rate of change of energy inside the CV)

$$\frac{\delta Q}{\delta t} - \frac{\delta W}{\delta t} = \frac{dE_{\text{sys}}}{dt} = \frac{d}{dx} \left(\int_{\text{CV}} e \rho dV \right) + \int_{\text{CS}} e \rho (\mathbf{V} \cdot \mathbf{n}) dA$$

Recall the sign convention: positive Q denotes heat added to the system
and positive W denotes work done by the system.

The Energy Equation

- The total system energy per unit mass e is expressed by

$$e = \underbrace{e_{\text{internal}}}_{\text{Microscopic}} + \underbrace{e_{\text{kinetic}} + e_{\text{potential}}}_{\text{Macroscopic}} = u + \frac{1}{2}V^2 + gz$$

- The work term is divided in three parts,

$$\dot{W} = \dot{W}_{\text{shaft}} + \dot{W}_{\text{press}} + \dot{W}_{\text{viscous}}$$

$$\text{with } \dot{W}_v \approx 0 \text{ and } \dot{W}_p = \int -p(-\mathbf{V} \cdot \mathbf{n})dA$$

$$\dot{W} = \dot{W}_s - \int p(-\mathbf{V} \cdot \mathbf{n})dA$$

(Shaft work is deliberately done by a machine, fan blade, piston, pump impeller through the control surface into the control volume)

- The heat transfer term is from conduction, radiation, and convection

The Energy Equation

- Recall energy equation,

$$\frac{\delta Q}{\delta t} - \frac{\delta W}{\delta t} = \frac{dE_{\text{sys}}}{dt} = \frac{d}{dt} \left(\int_{\text{CV}} e \rho dV \right) + \int_{\text{CS}} e \rho (\mathbf{V} \cdot \mathbf{n}) dA \quad \text{and} \quad \dot{W} = \dot{W}_s - \int p (-\mathbf{V} \cdot \mathbf{n}) dA$$

- Integrate using Reynolds transform theorem,

$$\dot{Q} - \dot{W}_s = \frac{\partial}{\partial t} \left(\int_{\text{CV}} e \rho dV \right) + \int_{\text{CS}} \left(e + \frac{p}{\rho} \right) \rho (\mathbf{V} \cdot \mathbf{n}) dA$$

- Energy equation for steady flow, 1-D inlets and outlets (1st law)

$$\dot{Q} - \dot{W}_s = \sum \dot{m}_{\text{out}} \left(u + \frac{1}{2} V^2 + gz + \frac{p}{\rho} \right)_{\text{out}} - \sum \dot{m}_{\text{in}} \left(u + \frac{1}{2} V^2 + gz + \frac{p}{\rho} \right)_{\text{in}}$$

The Energy Equation

- One inlet/outlet (1-inlet, 2-outlet, $\dot{m}_1 = \dot{m}_2 = \dot{m}_3$),

$$\dot{Q} - \dot{W}_s = \sum \dot{m}_{\text{out}} \left(u + \frac{1}{2} V^2 + gz + \frac{p}{\rho} \right)_{\text{out}} - \sum \dot{m}_{\text{in}} \left(u + \frac{1}{2} V^2 + gz + \frac{p}{\rho} \right)_{\text{in}}$$

$$\rightarrow \dot{Q} - \dot{W}_s = \dot{m} \left[(h_2 - h_1) + \frac{V_2^2 - V_1^2}{2} + g(z_2 - z_1) \right] \quad \text{where } h \equiv u + \frac{p}{\rho}$$

- Rearrange using the per unit mass form, (divide by $\dot{m}$)

$$q - w_s = h_2 - h_1 + \frac{V_2^2 - V_1^2}{2} + g(z_2 - z_1) \quad \text{where } q = \frac{\dot{Q}}{\dot{m}} \quad w_s = \frac{\dot{W}_s}{\dot{m}}$$

- Head form of the energy equation, (*pressure head* = $\frac{p}{\gamma} = \frac{p}{\rho g}$)

$$h_q - h_s = \left(\frac{u_2}{g} + \frac{p_2}{\gamma} + \frac{V_2^2}{2g} + z_2 \right) - \left(\frac{u_1}{g} + \frac{p_1}{\gamma} + \frac{V_1^2}{2g} + z_1 \right) \quad \text{where } h_q = \frac{q}{g} \quad h_s = \frac{w_s}{g}$$

The Energy Equation

- Energy eq. for steady, incompressible flow with one inlet/outlet,

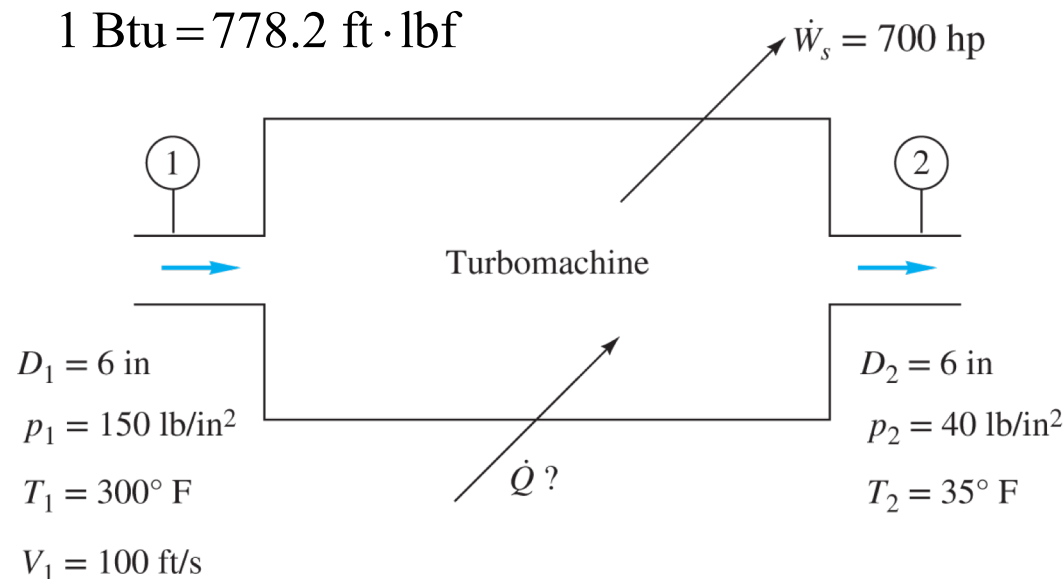
$$\left(\frac{p_1}{\gamma} + \frac{V_1^2}{2g} + z_1 \right)_{\text{in}} = \left(\frac{p_2}{\gamma} + \frac{V_2^2}{2g} + z_2 \right)_{\text{out}} + h_{\text{turbine}} - h_{\text{pump}} + h_f$$

Pressure head Velocity head Turbine head Pump head Head loss

where $h_f = \frac{(u_2 - u_1 - q)}{g}$

The Energy Equation

Example: Air ($R=1716, c_p=6003 \text{ ft} \cdot \text{lbf}/(\text{slug} \cdot ^\circ\text{R})$) flow steadily through a turbine that produce 700 hp of power. For the inlet and exit conditions shown, estimate (a) exit velocity V_2 and (b) the heat transfer rate in Btu/h.



The Energy Equation

Solution:

Select air in the turbine as **CV**, and assume air as ideal gas.

The inlet and exit densities,

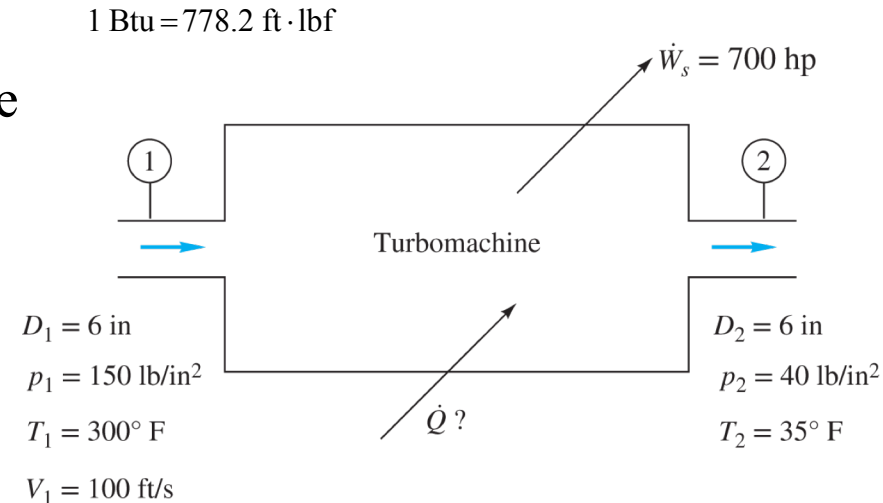
$$\rho_1 = \frac{p_1}{RT_1} = \frac{150(144)}{1716(300 + 460)} = 0.0166 \text{ slug/ft}^3$$

$$\rho_2 = \frac{p_2}{RT_2} = \frac{40(144)}{1716(35 + 460)} = 0.00679 \text{ slug/ft}^3$$

The mass flow rate is $\dot{m} = \rho_1 A_1 V_1 = (0.0166) \frac{\pi}{4} \left(\frac{6}{12} \right)^2 (100) = 0.325 \text{ slug/s}$

The exit velocity,

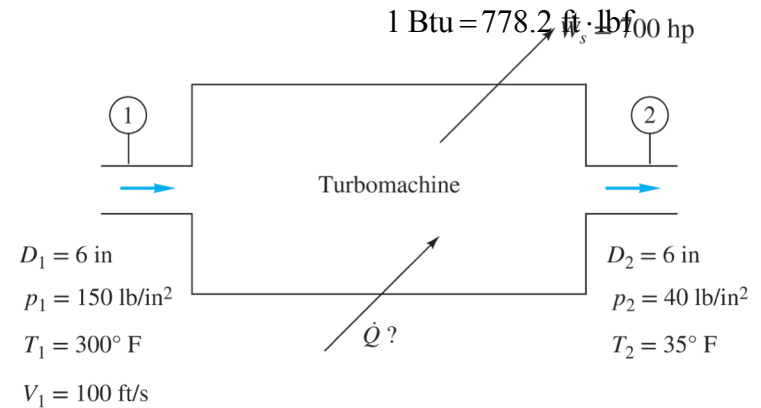
$$\dot{m} = 0.325 = \rho_2 A_2 V_2 = (0.00679) \frac{\pi}{4} \left(\frac{6}{12} \right)^2 V_2 \quad \rightarrow \quad V_2 = 244 \text{ ft/s}$$



The Energy Equation

$$\dot{m} = \rho_1 A_1 V_1 = (0.0166) \frac{\pi}{4} \left(\frac{6}{12} \right)^2 (100) = 0.325 \text{ slug/s}$$

$$V_2 = 244 \text{ ft/s}$$



$$\dot{Q} - \dot{W}_s = \sum \dot{m}_{\text{out}} \left(u + \frac{1}{2} V^2 + gz + \frac{p}{\rho} \right)_{\text{out}} - \sum \dot{m}_{\text{in}} \left(u + \frac{1}{2} V^2 + gz + \frac{p}{\rho} \right)_{\text{in}}$$

Steady flow energy equation applies with $z_1 = z_2$, $h = c_p T$,

$$\dot{Q} - \dot{W}_s = \dot{m} \left[c_p (T_2 - T_1) + \frac{V_2^2 - V_1^2}{2} \right]$$

$$\dot{Q} - 700(550) = 0.325 \left[6003(495 - 760) + \frac{244^2 - 100^2}{2} \right]$$

$$\dot{Q} = -125,000 \text{ ft} \cdot \text{lbf/s} = -578,000 \text{ Btu/h}$$

Summary

- All the laws of solid mechanics are written for a system, defined as an arbitrary quantity of mass of fixed identity.
- Reynolds transport theorem connects the mathematics between a system of individual masses and a region specified as the control volume (CV).

- The conservation of mass for steady and 1-D flow,

$$\sum (\rho_i A_i V_i)_{\text{out}} - \sum (\rho_i A_i V_i)_{\text{in}} = 0$$

- The linear momentum equation for steady and 1-D flow,

$$\sum \mathbf{F} = \sum_i (\dot{m}_i \mathbf{V}_i)_{\text{out}} - \sum_i (\dot{m}_i \mathbf{V}_i)_{\text{in}}$$

- Two main points for the CV analysis: 1. selection of a proper, clever, and workable CV; 2. reduction of the analysis that applies to the problem

- The energy eq. for steady, incompressible flow with one inlet/outlet

$$\left(\frac{p_1}{\gamma} + \frac{V_1^2}{2g} + z_1 \right)_{\text{in}} = \left(\frac{p_2}{\gamma} + \frac{V_2^2}{2g} + z_2 \right)_{\text{out}} + h_{\text{turbine}} - h_{\text{pump}} + h_f$$

- If the friction is negligible or tracing along a streamline in a fluid flow,

$$\frac{p_1}{\rho} + \frac{1}{2} V_1^2 + g z_1 = \frac{p_2}{\rho} + \frac{1}{2} V_2^2 + g z_2 = \text{const} \quad \text{Bernoulli equation}$$